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Integrals and sums

1.1 Multiline subscripts and superscripts

The `\substack` command can be used to produce a multiline subscript or superscript: for example

```
\sum_{\substack{
         0\le i\le m\\
         0<j<n}}
  P(i,j)
```

$$\sum_{\substack{0\le i\le m\\ 0<j<n}} P(i,j)$$

A slightly more generalized form is the `subarray` environment which allows you to specify that each line should be left-aligned instead of centered, as here:

```
\sum_{\begin{subarray}{l}
        i\in\Lambda\\ 0<j<n
      \end{subarray}}
 P(i,j)
```

$$\sum_{\substack{i\in\Lambda\\ 0<j<n}} P(i,j)$$

1.2 The `\sideset` command

There's also a command called `\sideset`, for a rather special purpose: putting symbols at the subscript and superscript corners of a large operator symbol such as $\sum$ or $\prod$. *Note: this command is not designed to be applied to anything other than sum-class symbols.* The prime example is the case when you want to put a prime on a sum symbol. If there are no limits above or below the sum, you could just use `\nolimits`: here's `\sum\nolimits' E_n` in display mode:

$$\sum\nolimits' E_n \tag{1.1}$$

If, however, you want not only the prime but also something below or above the sum symbol, it's not so easy—indeed, without `\sideset`, it would be downright difficult. With `\sideset`, you can write

```
\sideset{}{'}
  \sum_{n<k,\;\text{$n$ odd}} nE_n
```

$$\sum_{n<k,\ n\text{ odd}}' nE_n$$

The extra pair of empty braces is explained by the fact that `\sideset` has the capability of putting an extra symbol or symbols at each corner of a large operator; to put an asterisk at each corner of a product symbol, you would type

```
\sideset{_*^*}{_*^*}\prod
```

$$\sideset{_*^*}{_*^*}\prod$$

1.3 Placement of subscripts and limits

The default positioning for subscripts depends on the base symbol involved. The default for sum-class symbols is 'displaylimits' positioning: When a sum-class symbol appears in a displayed formula, subscript and superscript are placed in 'limits' position above and below, but in an inline formula, they are placed to the side, to avoid unsightly and wasteful spreading of the surrounding text lines. The default for integral-class symbols is to have sub- and superscripts always to the side, even in displayed formulas. (See the discussion of the `intlimits` and related options in Section **??**.)

Operator names such as sin or lim may have either 'displaylimits' or 'limits' positioning depending on how they were defined. The standard operator names are defined according to normal mathematical usage.

The commands `\limits` and `\nolimits` can be used to override the normal behavior of a base symbol:

$$\sum\nolimits_X, \qquad \iint\limits_A, \qquad \varinjlim\nolimits_{n\to\infty}$$

To define a command whose subscripts follow the same 'displaylimits' behavior as `\sum`, put `\displaylimits` at the tail end of the definition. When multiple instances of `\limits`, `\nolimits`, or `\displaylimits` occur consecutively, the last one takes precedence.

1.4 Multiple integral signs

`\iint`, `\iiint`, and `\iiiint` give multiple integral signs with the spacing between them nicely adjusted, in both text and display style. `\idotsint` is an extension of the same idea that gives two integral signs with dots between them.

$$\iint\limits_A f(x,y)\,dx\,dy \qquad \iiint\limits_A f(x,y,z)\,dx\,dy\,dz \tag{1.2}$$

$$\iiiint\limits_A f(w,x,y,z)\,dw\,dx\,dy\,dz \qquad \int\limits_A \cdots \int f(x_1,\ldots,x_k) \tag{1.3}$$

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Commutative diagrams

Some commutative diagram commands like the ones in $\mathcal{A}\mathcal{M}\mathcal{S}$-TEX are available as a separate package, `amscd`. For complex commutative diagrams authors will need to turn to more comprehensive packages like Ti*k*Z (in particular, `tikz-cd`) or XY-pic, but for simple diagrams without diagonal arrows the `amscd`

commands may be more convenient. Here is one example.

$$\begin{array}{ccc} S^{\mathcal{W}_\Lambda}\otimes T & \xrightarrow{\ j\ } & T \\ \Big\downarrow & & \Big\downarrow{\scriptstyle \mathrm{End}\ P} \\ (S\otimes T)/I & = & (Z\otimes T)/J \end{array}$$

```
\begin{CD}
S^{\mathcal{W}_\Lambda}\otimes T   @>j>>   T\\
@VVV                                      @VV{\End P}V\\
(S\otimes T)/I                  @=        (Z\otimes T)/J
\end{CD}
```

In the `CD` environment the commands `@>>>`, `@<<<`, `@VVV`, and `@AAA` give respectively right, left, down, and up arrows. For the horizontal arrows, material between the first and second `>` or `<` symbols will be typeset as a superscript, and material between the second and third will be typeset as a subscript. Similarly, material between the first and second or second and third `A`s or `V`s of vertical arrows will be typeset as left or right "sidescripts". The commands `@=` and `@|` give horizontal and vertical double lines. A "null arrow" command `@.` can be used instead of a visible arrow to fill out an array where needed.

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Using math fonts

3.1 Introduction

For more comprehensive information on font use in LaTeX, see the LaTeX font guide (`fntguide.tex`) or *The LaTeX Companion* [**?**]. The basic set of math font commands in LaTeX includes `\mathbf`, `\mathrm`, `\mathcal`, `\mathsf`, `\mathtt`, `\mathit`. Additional math alphabet commands such as `\mathbb` for blackboard bold, `\mathfrak` for Fraktur, and `\mathscr` for Euler script are available through the packages `amsfonts` and `euscript` (distributed separately).

3.2 Recommended use of math font commands

If you find yourself employing math font commands frequently in your document, you might wish that they had shorter names, such as `\mb` instead of `\mathbf`. Of course, there is nothing to keep you from providing such abbreviations for yourself by suitable `\newcommand` statements. But for LaTeX to provide shorter names would actually be a disservice to authors, as that would obscure a much better alternative: defining custom command names derived from the names of the underlying mathematical objects, rather than from the names of the fonts used to distinguish the objects. For example, if you are using bold to indicate vectors, then you will be better served in the long run if you define a 'vector' command instead of a 'math-bold' command:

```
\newcommand{\vect}[1]{\mathbf{#1}}
```

you can write `\vect{a} + \vect{b}` to produce $\mathbf{a} + \mathbf{b}$. If you decide several months down the road that you want to use the bold font for some other purpose, and mark vectors by a small over-arrow instead, then you can put the change into effect merely by changing the definition of `\vect`; otherwise you would have to replace all occurrences of `\mathbf` throughout your document, perhaps even needing to inspect each one to see whether it is indeed an instance of a vector.

It can also be useful to assign distinct command names for different letters of a particular font:

```
\DeclareSymbolFont{AMSb}{U}{msb}{m}{n}% or use amsfonts package
\DeclareMathSymbol{\C}{\mathalpha}{AMSb}{"43}
\DeclareMathSymbol{\R}{\mathalpha}{AMSb}{"52}
```

These statements would define the commands `\C` and `\R` to produce blackboard-bold letters from the 'AMSb' math symbols font. If you refer often to the complex numbers or real numbers in your document, you might find this method more convenient than (let's say) defining a `\field` command and writing `\field{C}`, `\field{R}`. But for maximum flexibility and control, define such a `\field` command and then define `\C` and `\R` in terms of that command:

```
\usepackage{amsfonts}% to get the \mathbb alphabet
\newcommand{\field}[1]{\mathbb{#1}}
\newcommand{\C}{\field{C}}
\newcommand{\R}{\field{R}}
```

3.3 Bold math symbols

The `\mathbf` command is commonly used to obtain bold Latin letters in math, but for most other kinds of math symbols it has no effect, or its effects depend unreliably on the set of math fonts that are in use. For example, writing

```
\Delta \mathbf{\Delta}\mathbf{+}\delta \mathbf{\delta}
```

produces $\Delta\mathbf{\Delta}+\delta\mathbf{\delta}$; the `\mathbf` has no effect on the plus sign or the small delta. The amsmath package therefore provides two additional commands, `\boldsymbol` and `\pmb`, that can be applied to other kinds of math symbols. `\boldsymbol` can be used for a math symbol that remains unaffected by `\mathbf` if (and only if) your current math font set includes a bold version of that symbol. `\pmb` can be used as a last resort for any math symbols that do not have a true bold version provided by your set of math fonts; "pmb" stands for "poor man's bold" and the command works by typesetting multiple copies of the symbol with slight offsets. The quality of the output is inferior, especially for symbols that contain any hairline strokes. When the standard default set of LaTeX math fonts are in use (Computer Modern), the only symbols that are likely to require `\pmb` are large operator symbols like `\sum`, extended delimiter symbols, or the extra math symbols provided by the `amssymb` package [**?**].

The following formula shows some of the results that are possible:

```
A_\infty + \pi A_0
\sim \mathbf{A}_{\boldsymbol{\infty}} \boldsymbol{+}
  \boldsymbol{\pi} \mathbf{A}_{\boldsymbol{0}}
\sim\pmb{A}_{\pmb{\infty}} \pmb{+}\pmb{\pi} \pmb{A}_{\pmb{0}}
```

$$A_\infty + \pi A_0 \sim \mathbf{A}_{\boldsymbol{\infty}} \boldsymbol{+} \boldsymbol{\pi} \mathbf{A}_{\boldsymbol{0}} \sim \boldsymbol{A}_{\boldsymbol{\infty}} \boldsymbol{+} \boldsymbol{\pi} \boldsymbol{A}_{\boldsymbol{0}}$$

If you want to use only the `\boldsymbol` command without loading the whole amsmath package, the `bm` package is recommended (this is a standard LaTeX package, not an AMS package; you probably have it already if you have a 1997 or newer version of LaTeX).

3.4 Italic Greek letters

For italic versions of the capital Greek letters, the following commands are provided:

`\varGamma`	$\varGamma$	`\varSigma`	$\varSigma$
`\varDelta`	$\varDelta$	`\varUpsilon`	$\varUpsilon$
`\varTheta`	$\varTheta$	`\varPhi`	$\varPhi$
`\varLambda`	$\varLambda$	`\varPsi`	$\varPsi$
`\varXi`	$\varXi$	`\varOmega`	$\varOmega$
`\varPi`	$\varPi$		

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Error messages and output problems

4.1 General remarks

This is a supplement to Chapter 8 of the LaTeX manual [**?**] (first edition: Chapter 6). Appendix B of the *Companion* [**?**] lists all LaTeX errors including those for amsmath with likely causes for each. For the reader's convenience, the set of error messages discussed here overlaps somewhat with the set in those references, but please be aware that we don't provide exhaustive coverage here. The error messages are arranged in alphabetical order, disregarding unimportant text such as `! LaTeX Error:` at the beginning, and nonalphabetical characters such as `\`. Where examples are given, we show also the help messages that appear on screen when you respond to an error message prompt by entering `h`.

Remember that the important error message is the first for a given line. When running in nonstop mode, errors accumulate, and the first error may create a further error condition from which it is impossible to recover. In such a situation processing will stop after 100 errors, so the reporting of genuine errors may not be complete, and it may be impossible to determine which reported errors are genuine and which are not.

A final section discusses some output errors, i.e., instances where the printed document has something wrong but there was no LaTeX error during typesetting.